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Title: Meningoencephalitis Guidelines

Introduction:

Meningitis is inflammation of meninges. It is caused by infectious and non-infectious causes. Infectious causes are bacterial, viral, Tuberculosis, parasitic, fungi and amebic meningitis. Bacterial meningitis remains a disease with high mortality and long-term morbidity. 90% of bacterial meningitis cases are caused by *Streptococcus pneumoniae* .This guideline is going to concentrate mainly on management of bacterial meningitis.

Objectives:

-To provide the clinicians with initial management approach to the patients with suspected acute bacterial meningitis with or without encephalitis. It focus on early recognition of the meningitis, rapid diagnostic evaluation, and emergent antimicrobial and adjunctive therapy of bacterial meningitis.

Signs and symptoms:

Meningitis:

Meningitis is an inflammation of the meninges surrounding the brain and spinal cord.

Unlike adult patients who usually present with fever, headache, meningeal signs ± photophobia± altered mental status; children usually present with fever, poor oral intake, vomiting, and lethargy or irritability ± seizure.

Encephalitis:

Inflammation of the cerebral cortex. The patients present with fever, headache & altered mental status, behavioral, personality and speech problem.Obtundation and coma. Encephalitis most commonly caused by viruses. Difference from meningitis more likely with personality and behavioral changes.

Common bacterial cause of acute bacterial meningitis and Antimicrobial therapy

Meningitis	Neonate	Group B <i>Streptococcus</i> , gram negative enteric bacilli (e.g. <i>E. Coli</i>), <i>Listeria monocytogenes</i> (rare)	Cefotaxime ± Ampicillin
Meningitis	1-3 months	Includes organisms usually seen in neonates or older children	Cefotaxime + Vancomycin ± Ampicillin
Meningitis	Older Children	<i>S. Pneumonia</i> , <i>N. meningitidis</i> , <i>H. Influenzae</i>	Ceftriaxone + Vancomycin
Brain abscess	<i>Strept. spp.</i> , <i>Bacteroides</i> , Enterobacteriaceae, <i>S. milleri</i> , <i>S. aureus</i> , Rare: <i>Nocardia</i> , <i>Listeria</i> , Consider <i>Toxoplasma gondii</i> in HIV patient	Ceftriaxone + Vancomycin + Metronidazole Consult Infectious diseases specialist	Use initially Vancomycin to cover MRSA or beta lactam resistant <i>S. pneumoniae</i> then adjusts according to culture results. Use Ceftriaxone instead of Ceftriaxone if secondary to chronic otitis externa. Meropenem may be used in situation of antibiotic resistant (ESBL) <i>S. aureus</i> more likely if endocarditis +ve blood culture. Need neurosurgical consult for potential drainage

CNS Infection	Organisms	Antimicrobial Agent
Shunt Infection	Coagulase-negative, <i>Staphylococcus</i> , coliform bacilli, <i>Pseudomonas</i>	Vancomycin + Ceftriaxone Or Vancomycin + Cefazidime Or Cefepime or Meropenem
Subdural empyema	Most of the time its extension of sinusitis or otitis media	Same as for primary brain abscess Surgical emergency and must be drained.
Encephalitis:	Herpes simplex	IV : Acyclovir 20mg/kg/dose every 8hrs and for 21 days

Indication of CT head:

- 1-A history of CNS disease (e.g., mass lesion, stroke and focal infection), an immunocompromised state (e.g., that due to HIV infection or AIDS, immunosuppressive therapy, transplantation).
- 2-History of prolonged seizure.
- 3-Certain specific abnormal neurologic findings (e.g. an abnormal level of consciousness, an inability to answer 2 consecutive questions correctly or to follow 2 consecutive commands, gaze palsy, abnormal visual fields, facial palsy, arm drift, leg drift, abnormal language).
- 4-Signs of increase intracranial pressure like papilledema advise to treat high intracranial pressure then to do CT and LP if possible.

Lumbar puncture contraindication:

- Skin infection near the site of the lumbar puncture.
- Suspicion of increased intracranial pressure due to a cerebral mass.
- Uncorrected coagulopathy
- Acute spinal cord trauma.
- Markedly reduced GCS
- Cardiovascular compromise

How to differentiate between bacterial and viral meningitis?

- 1-CSF laboratory test: (microscopy, biochemistry, culture and Viral PCR)

meningitis.

Gram stain of CSF: sensitivity of 60%–90% of patients with community-acquired bacterial meningitis & specificity of 97%.

Latex agglutination. In general a sensitivity of 70% and specificity of 99.4%.

Culture: sensitivity 43% and specificity 97 %⁽²⁾.

Viral PCR: sensitivity 60% and specificity 97 %⁽²⁾.

CSF bacterial PCR: could be considered in case of negative culture or undertreated or partially treated meningitis.

Indications for repeating Lumbar Puncture in patients with meningitis.

1-If the patient doesn't response clinically after 48 h of appropriate antimicrobial therapy.

2-The neonate with meningitis due to gram-negative bacilli should undergo repeated lumbar punctures to document CSF sterilization, because the duration of antimicrobial therapy is determined, in part, by the result.

3-In patients with CSF shunt infections, the presence of drainage catheter after shunt removal allows for monitoring of CSF parameters to ensure that the infection is responding to appropriate antimicrobial therapy and drainage.

Antimicrobial therapy doses:

i. Children antimicrobial doses:

- **Vancomycin:** 15 mg/kg/dose q6h for normal renal function test.
- **Ceftriaxone:** 50 mg/kg/dose q12or
- **Cefotaxime:** 50 mg/kg/dose q6h.
- **Ampicillin:** 100mg/kg/dose q6h.
- **Penicillin G >28 days:** 300,000 units/kg/day q4h.
- **Acyclovir for HSV encephalitis.**
- <3 months: 20 mg/kg/dose q8h.
- >3 month: 15mg/kg/dose q8h.

Role of dexamethasone therapy in-patients with bacterial meningitis?

Neonates : Steroids aren't encouraged in neonates due to feasible results on neurodevelopment

-Infants and children: At present, there are mixed data to make a recommendation on the use of adjunctive dexamethasone in neonates with bacterial meningitis.

- If giving the first dose of IV dexamethasone after initial antibiotic management, this ought to preferably be carried out within four hours and not extra than 12 hours after starting antibiotics

Duration of Antimicrobial Therapy:

Duration of antimicrobial therapy for bacterial meningitis based on isolated microorganism

Micro organism	Duration of therapy, days
<i>Neisseria meningitides</i>	7
<i>Haemophilus influenza</i>	10
<i>Streptococcus pneumoniae</i>	10-14
<i>Streptococcus agalactiae</i>	14-21
<i>Staph. Aureus</i>	14
<i>Aerobic gram-negative bacilli</i>	21
<i>Listeria monocytogenes</i>	21

In patients who develop bacterial meningitis after placement of CSF Shunt, Is it necessary to administer antimicrobial therapy by the intraventricular Route?

Direct instillation of antimicrobial agents into the ventricles through either an external ventriculostomy or shunt reservoir is occasionally necessary in patients who have shunt infections that are difficult to eradicate or who cannot undergo the surgical components of therapy.

In patients with shunts infection does the shunt need to be removed for optimal therapy, and when can a new shunt be implanted?

hydrocephalus until the infection has cleared.

Recommended dosages of antimicrobial agents administered by the intraventricular route:

Antimicrobial agent, daily, intraventricular dose in mg:

Antimicrobial agent	Intraventricular dose in mg/day (not per kg)
Vancomycin	5-20
Gentamycin	1-8
Tobramycin	5-20
Amikacin	5-50
Polymyxin B	5
Colistin	10
Quinupristin/dalfopristin	2-5
Teicoplanin	5

When antimicrobial therapy is administered via a ventricular drain, the drain should be clamped for 15–60 minutes to allow the agent to equilibrate throughout the CSF.

In patients with repeatedly positive CSF cultures on appropriate antimicrobial therapy, treatment should be continued for 10–14 after the last positive culture.

Re-shunting:

1-Coagulase-negative staphylococci (CONS) shunt infection

1-If the patient has normal CSF findings & negative CSF culture results after externalization and shunt removal can be reshunted on the third day after removal.

2-If CSF abnormalities are present & CONS is isolated, 7 days of antimicrobial therapy are recommended prior to reshunting as long as additional CSF culture results are negative and the

ventricular protein concentration is appropriate (200 mg/dL); if additional culture results are positive, antimicrobial therapy is continued until CSF culture results remain negative for 10

Consecutive days before a new CSF shunt is placed.

2-*S. aureus*, shunt infections

10 days of negative culture results are recommended prior to reshunting

3-Gram-negative bacilli

10–14-day course of antimicrobial therapy should be used, although longer durations may be needed depending on the clinical response. Some experts also suggest that consideration be given to a 3-day period off antimicrobial therapy to verify clearing of the infection prior to shunt reimplantation; although this approach is optional, it may not be necessary for all patients.

Empirical Therapy

Acyclovir should be initiated in all patients with suspected Encephalitis (those with altered mental status, acute cognitive dysfunction, behavioral changes, focal neurologic signs, and prolonged seizures. In most cases, there is some concomitant meningeal inflammation.

Specific Therapy

Viruses

- Herpes simplex or Varicella-zoster virus: acyclovir is recommended.
- Epstein-Barr virus, Enteroviruses & mumps antiviral medication is not recommended;
- Human herpesvirus 6: ganciclovir or foscarnet should be used in immunocompromised patients
- Measles virus: ribavirin can be considered, intrathecal ribavirin can be considered in patients with subacute sclerosing panencephalitis.

What neurodiagnostic tests should be performed in patients with suspected encephalitis?

With MRI being more sensitive and specific.

What specific criteria should be used for day care antimicrobial therapy in the patient with bacterial meningitis?

- Inpatient antimicrobial therapy for ≥ 6 days,

-No significant neurological deficit, focal finding or seizure.

-Clinical stability or improving condition.

-Ability to take fluids by mouth.

-Reliable IV access.

-Established management plan

-Safe environment with easy access to the patient.

Notification

- All cases of presumed or confirmed *N meningitidis*, *Pneumococcal* meningitis and HiB should be notified

- All cases of Acute encephalitis syndrome should be notified.

Complications

Hearing impairment

Neurodevelopmental impairment

Multiorgan involvement due to disseminated infection

Venous sinus thrombosis

Seizures, subsequent epilepsy

Permanent focal neurological deficit

Hydrocephalus

Follow up:

- All children with encephalitis or bacterial meningitis need to have a formal audiology assessment 6–8 weeks after discharge (in advance if concerns)

-Neurodevelopmental progress ought to be monitored in outpatients

- Consider investigating for complement deficiency if the child has had >1 episode of meningococcal disease

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